GENERAL CARBON THEORY FOR FLEXIBLE COMPUTING LOADS: THE CAPACITY-SUBSTITUTION CHANNEL AND A DISTRIBUTION-FREE NET-CARBON-REDUCTION CONDITION

TEAM A1 — ROLE 1 (GENERAL CARBON THEORY)

ABSTRACT. We harden and generalize the carbon analysis of the flexible-computing-load (FCL) model beyond the uniform-distribution benchmark. Four results. *First*, we re-derive the uniform closed forms (paper eqs. 18–26) from the first-order conditions and verify them to machine precision against direct quadrature; along the way we establish the *knife-edge* structure for the entire power-law family and reconcile two transcription issues in the current draft. *Second*, we prove the central comparative-statics claim $K_R^* > K_R^\circ$—renewable capacity is strictly higher with the FCL program—for *any* residual-demand distribution satisfying a single, economically transparent condition on the *shift ratio* $g(x) = F_{K_R}(x)/f(x)$: the program raises renewable capacity whenever an extra megawatt of renewables relaxes residual demand more in deep-surplus states than in tight states. The power-law family satisfies this for every shape parameter, which is exactly why footnote 6's extension goes through. *Third*, we give the exact net-carbon inequality, decompose it into a distribution-free *utilization* effect and a sign-controlled *capacity-substitution* effect, and state a clean sufficient condition for the carbon part of the triple win. *Fourth*, we give a distribution-free reading of the effect: the carbon change is governed by a *two-moment sufficient statistic*, bracketed by closed-form *magnitude bounds* computable from cutoff data alone, and—over the natural shape class—worst-cased by the *uniform*, so the closed-form condition is a distributionally robust certificate. Every analytical formula is verified to machine precision against finite differences of the profit integral. The four points left open by the uniform-only treatment—the lattice (submodularity) obstruction, the endogenous-capacity feedback in the carbon inequality, an explicit renewable-cost threshold, and the regime boundaries—are each closed here; a companion memo keeps the same accounting.

1. Setup and notation

We work with the model of Section 3 of the paper and adopt its notation. A regulated utility chooses renewable capacity K_R, dispatchable fossil backup K_F, and flexible-computing-load capacity K_M to maximize expected profit over one representative period. The state variable is *residual demand*

$$\tilde{x} := \tilde{D} - K_R \tilde{S} \in [-K_R, \bar{D}], \tag{1}$$

where $\tilde{D} \in [0, \bar{D}]$ is stochastic consumer demand and $\tilde{S} \in [0,1]$ the renewable capacity factor. Let $F(x; K_R)$ be the CDF of $\tilde{x}$ with density $f(x; K_R)$, extended by $F \equiv 0$ for $x < -K_R$ and $F \equiv 1$ for $x > \bar{D}$. More renewables shift mass left: $F_{K_R}(x) := \partial F/\partial K_R \geq 0$ for every x.

Per-MWh quantities: FCL revenue R, fossil variable cost c, retail price p, blackout penalty β. Amortized per-MW capital costs: C_R, C_F, C_M. We maintain the paper's parameter regularity.

Assumption 1 (Regularity)**.** $R > c + C_M, \quad R < c + C_F + C_M, \quad \beta + p > c + C_F, \quad \beta + p > R.$

Under the optimal within-period dispatch rule $d(x) = \min\{\max\{K_F - x, 0\}, K_M\}$ (paper Lemma 1), plugging into expected profit and integrating by parts gives the unified profit

$$\Pi(K_R, K_F, K_M) = p\,\mathbb{E}[\tilde{D}] + R\int_{K_F - K_M}^{K_F} F\,dx - c\int_{-K_M}^{K_F - K_M} (1-F)\,dx - (\beta+p)\int_{K_F}^{\bar{D}} (1-F)\,dx - C_F K_F - C_R K_R - C_M K_M, \tag{2}$$

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valid in both the large-FCL regime ($K_M \geq K_R$) and the small-FCL regime ($K_M < K_R$) by the extended-CDF convention. Setting $K_M = 0$ gives the no-FCL benchmark Π_0.

Lemma 1 (Unified first-order conditions)**.** *Differentiating (2) by Leibniz's rule,*

$$\frac{\partial \Pi}{\partial K_M} = (R-c)\,F(K_F - K_M; K_R) + c\,F(-K_M; K_R) - C_M, \tag{3}$$

$$\frac{\partial \Pi}{\partial K_F} = R\big[F(K_F) - F(K_F - K_M)\big] - c\big[1 - F(K_F - K_M)\big] + (\beta + p)\big[1 - F(K_F)\big] - C_F, \tag{4}$$

$$\frac{\partial \Pi}{\partial K_R} = R\int_{K_F - K_M}^{K_F} F_{K_R}\,dx + c\int_{-K_M}^{K_F - K_M} F_{K_R}\,dx + (\beta + p)\int_{K_F}^{\bar{D}} F_{K_R}\,dx - C_R. \tag{5}$$

In the large-FCL regime $F(-K_M; K_R) = 0$, and (3)–(5) collapse to the paper's eqs. (12), (13), (15) respectively.[1]

Lemma 2 (Distribution-free equilibrium probabilities)**.** *At any interior optimum, the CDF values at the equilibrium thresholds are* independent of the shape of F*:*

$$F(K_F^\circ; K_R^\circ) = \frac{\beta + p - c - C_F}{\beta + p - c} := \varphi^\circ, \qquad F(K_F^*; K_R^*) = \frac{\beta + p - c - C_F - C_M}{\beta + p - R}, \qquad F(K_F^* - K_M^*; K_R^*) = \frac{C_M}{R - c}. \tag{6}$$

Proof. The first identity is the no-FCL FOC (4) with $K_M = 0$; the other two solve (3)–(4). Each fixes a *value* of F, not a capacity, so no distributional assumption enters. *(Verified on the non-uniform* Beta$(2,2)$ *scale-mixture: both equal* $\frac{5}{9}$ *and* $\frac{17}{18}$ *to* 10^{-9}*.)* □

These give the adoption threshold

$$\bar{C}_M = (R-c)\,\varphi^\circ = \frac{(R-c)(\beta + p - c - C_F)}{\beta + p - c}, \tag{7}$$

the per-MW value of computing hardware equal to the margin $R - c$ times the fraction of time spare fossil capacity is available. Lemma 2 is the lever that makes the carbon comparison of §4 general: the program shifts the equilibrium *capacities*, but the *probabilities* at which fossil cuts in and FCL saturates are pinned.

2. Ground truth: the uniform closed form

We re-derive the uniform benchmark used to sign the carbon result, both to lock in correctness and to fix the algebraic constants the general theory must reproduce. Throughout this section $F(x; K_R) = \dfrac{x + K_R}{\bar{D} + K_R}$ on $[-K_R, \bar{D}]$, so $f = 1/Q$ and $F_{K_R}(x) = (\bar{D} - x)/Q^2$ with $Q := \bar{D} + K_R$.

Lemma 3 (Uniform optimal capacities)**.** *Write $Q^\circ = \bar{D} + K_R^\circ$ and $Q^* = \bar{D} + K_R^*$. Then*

$$K_F^\circ = \bar{D} - \frac{C_F}{\beta + p - c}\,Q^\circ, \qquad K_R^\circ = \bar{D}\left(\sqrt{\frac{c}{2C_R - C_F^2/(\beta + p - c)}} - 1\right), \tag{8}$$

$$K_F^* = \bar{D} + \frac{R - c - C_F - C_M}{\beta + p - R}\,Q^*, \qquad K_F^* - K_M^* = \bar{D} - \frac{R - C_M - c}{R - c}\,Q^*. \tag{9}$$

Proof. (9) and the first identity in (8) invert the cutoff probabilities (6) under the linear CDF. For K_R°, substitute K_F° into the renewable FOC (5) with $K_M = 0$. Writing $\gamma := C_F/(\beta + p - c)$ so that

[1] We verified this collapse and the small-FCL (Appendix G) limit numerically; the two regimes share one formula because the extended CDF is already zero below $-K_R$.

$K_F^\circ = \bar{D} - \gamma Q^\circ$ and $\bar{D} - K_F^\circ = \gamma Q^\circ$, and using $\int_a^b F_{K_R}\,dx = [(b-a)(\bar{D} - \frac{a+b}{2})]/Q^2$,

$$\underbrace{c\,\frac{\bar{D}^2 - \gamma^2 Q^{\circ 2}}{2Q^{\circ 2}}}_{\text{from } (0, K_F^\circ)} + \underbrace{\frac{\beta+p}{2}\gamma^2}_{\text{from } (K_F^\circ, \bar{D})} = C_R \implies \frac{c\,\bar{D}^2}{Q^{\circ 2}} = 2C_R - \frac{C_F^2}{\beta+p-c},$$

because $\gamma^2(\beta+p-c) = C_F^2/(\beta+p-c)$. Solving for Q° and subtracting $\bar{D}$ gives (8). *(Verified to machine precision against the quadrature solver.)* □

Carbon (paper eqs. 25–26). Expected fossil output without the program is $\mathbb{E}[\text{fossil}^\circ] = K_F^\circ - \int_0^{K_F^\circ} F\,dx = K_F^\circ(\bar{D} - \frac{K_F^\circ}{2})/Q^\circ$, and with the program (large-FCL regime), accounting for the flat stretch $(K_M^* - K_R^*)$ on which the extended CDF equals zero,

$$\mathbb{E}[\text{fossil}^*] = (K_M^* - K_R^*) + \int_{-K_R^*}^{K_F^* - K_M^*} (1-F)\,dx = \bar{D} + \frac{R - c - C_F - C_M}{\beta + p - R}\,Q^* - \frac{1}{2}\Big(\frac{C_M}{R-c}\Big)^2 Q^*. \quad (10)$$

Carbon emissions equal c times these quantities. Net carbon falls with the program iff $\mathbb{E}[\text{fossil}^*] < \mathbb{E}[\text{fossil}^\circ]$.

Remark 1 (Two corrections to the current draft). *(i)* The bracketed expressions printed in eqs. (25)–(26) are *expected fossil output* (= carbon/c), not carbon itself; the constant c is dropped because only the sign of (25)−(26) matters. Restoring c, our (10) reproduces the printed eq. (26) to 6×10^{-10} and eq. (25) to machine precision. *(ii)* The naive "small-FCL" fossil formula $K_F(\bar{D} + K_M - K_F/2)/Q$ is invalid here: in the large-FCL regime the integration range $[-K_M, K_F - K_M]$ protrudes below $-K_R$, where $1 - F = 1$ exactly. The flat term $(K_M^* - K_R^*)$ in (10) is the correction.

Proposition 1 (Knife-edge for the power-law family)**.** *Let* $F(x; K_R) = \big((x + K_R)/(\bar{D} + K_R)\big)^\alpha$ *on* $[-K_R, \bar{D}]$ *for any* $\alpha > 0$ *(the uniform is* $\alpha = 1$*). Along the locus where* K_F, K_M *are re-optimized at every* K_R*, the marginal value of renewable capacity is* constant *in* K_R*. In the interior regime it equals*

$$\text{MV} = \tfrac{1}{2}\left[c + \frac{(R - C_M - c)^2}{R - c} + \frac{(R - c - C_F - C_M)^2}{\beta + p - R}\right], \quad (11)$$

so $K_R^* = \infty$ *if* $\text{MV} > C_R$*,* $K_R^* = 0$ *if* $\text{MV} < C_R$*, and any* K_R *is optimal at equality. Equivalently* $K_R^* = \infty$ *iff*

$$c + \frac{(R - C_M - c)^2}{R - c} + \frac{(R - c - C_F - C_M)^2}{\beta + p - R} > 2C_R, \quad (12)$$

which is exactly *the bracket printed in paper eq. (24).*

Proof. For the power law one computes $F_{K_R}(x) = f(x)\,(\bar{D} - x)/Q$, hence the shift ratio $g(x) = F_{K_R}/f = (\bar{D} - x)/Q$ is the *same* affine function for every α. Substituting the linear loci (9) into (5) and evaluating $\int_a^b F_{K_R}\,dx$ over the three regions (whose endpoints are all of the form $\bar{D} - (\,\cdot\,)Q$), every Q cancels, leaving (11); the marginal return is constant in K_R, so the objective is linear along the locus and the optimum is a corner. Multiplying (11) by 2 and subtracting $2C_R$ gives precisely the paper's eq. (24) bracket, confirming (eq. 24) $= 2\,(\text{MV} - C_R) = 2\,\partial\Pi/\partial K_R$ along the locus. *(MV constant in* K_R *verified at* $K_R \in \{1, 5, 9\}$ *for* $\alpha \in \{0.5, 1, 2\}$*.)* □

Remark 2 (On eq. (24)). Proposition 1 shows eq. (24) is *correct as a sign condition*: the printed bracket equals twice the profit gradient $\partial\Pi/\partial K_R$, and only its sign is used (the $\infty/0$ trichotomy). The factor of two is a grouping artifact of the closed-form substitution, not an error—worth a one-line clarifying footnote in the draft.

3. The capacity-substitution channel, in general

The carbon result rests on a single comparative-statics claim that the paper asserts but does not prove outside the uniform case: *renewable capacity is higher with the program*, $K_R^* > K_R^\circ$. We now prove it for any distribution satisfying one transparent condition. The obstruction, and why the naive lattice argument fails, is recorded in Section 8.

3.1. The marginal-value cross-partials.

From Lemma 1, the second derivatives of Π that govern how the optimum moves when computing becomes more viable are

$$\frac{\partial^2 \Pi}{\partial K_R \, \partial K_M} = (R - c)\, F_{K_R}(K_F - K_M) + c\, F_{K_R}(-K_M) \;\geq 0, \tag{13}$$

$$\frac{\partial^2 \Pi}{\partial K_F \, \partial K_M} = (R - c)\, f(K_F - K_M) \;\geq 0, \tag{14}$$

$$\frac{\partial^2 \Pi}{\partial K_R \, \partial K_F} = (R - \beta + p)\, F_{K_R}(K_F) - (R - c)\, F_{K_R}(K_F - K_M) \;\leq 0. \tag{15}$$

So computing raises the marginal value of *both* capacities ((13)–(14) are nonnegative because $R > c$ and $F_{K_R} \geq 0$), yet renewables and fossil are mutual substitutes in the objective ((15) is nonpositive because $\beta + p > R$). The sign conflict in (15) is exactly why a one-line Topkis argument does not deliver $K_R^* > K_R^\circ$; see Section 8.

3.2. The shift-ratio condition.

Define the *shift ratio*

$$g(x; K_R) := \frac{F_{K_R}(x; K_R)}{f(x; K_R)} = -\left.\frac{\partial x}{\partial K_R}\right|_{F \text{ fixed}}, \tag{16}$$

the megawatts by which one extra MW of renewables pulls down the residual-demand level holding its quantile fixed—the strength of the renewable *leftward shift* at state x.

Condition 1 (Shift ratio)**.** $g(x; K_R)$ is non-increasing in x on $(K_F^* - K_M^*,\, K_F^*)$ at each point of the equilibrium path $C_M \in [C_M, \bar{C}_M]$.

Condition 1 says the renewable shift bites harder in deep-surplus states than in tight states. It is mild and interpretable, and it holds for the entire power-law family, where $g(x) = (\bar{D} - x)/Q$ is affine and strictly decreasing for every α.

Condition 2 (Regularity)**.** On the adoption path $C_M \in [C_M, \bar{C}_M]$ the program has an interior maximizer (K_R^*, K_F^*, K_M^*) at which the Hessian $\mathcal{H}$ of Π is nonsingular.

Condition 2 is exactly what the implicit function theorem needs to make the path $C_M \mapsto (K_R^*, K_F^*, K_M^*)$ a C^1 curve; at a maximizer, nonsingularity forces $\mathcal{H} \prec 0$. It is verified directly: §7 computes the eigenvalues of $\mathcal{H}$ along the entire path and finds all three strictly negative throughout the interior region. The one place it fails is the *knife-edge boundary*, where C_R is small enough that $K_R^* \to \infty$ (Proposition 1): there $\mathcal{H}$ acquires a zero eigenvalue as the marginal value of renewables turns flat. But that boundary needs no comparative-statics argument—$K_R^* = \infty > K_R^\circ$ holds trivially—so Condition 2 costs nothing where Theorem 1 is actually used (interior optima with finite K_R^*).

Theorem 1 (Capacity-substitution channel)**.** *Under Conditions 1–2, at every interior point of the adoption path the equilibrium derivative is*

$$\frac{dK_R^*}{dC_M} = \frac{\operatorname{cof}_{K_M K_R}(\mathcal{H})}{\det \mathcal{H}} \;\leq 0, \qquad \operatorname{cof}_{K_M K_R}(\mathcal{H}) \xrightarrow[\text{large-FCL}]{} (R-\beta+p)(R-c)\big[F_{K_R}(K_F^*) f(K_F^* - K_M^*) - F_{K_R}(K_F^* - K_M^*) f(K \tag{17}$$

where $\mathcal{H}$ is the Hessian of Π (so $\det\mathcal{H}<0$), and the displayed bracket is the large-FCL ($K_M^*>K_R^*$) form of the cofactor. Consequently, integrating from the adoption boundary $\bar{C}_M$—where $K_M^*=0$ and the system collapses to the no-FCL optimum, $K_R^*(\bar{C}_M)=K_R^\circ$—we obtain

$$K_R^*(C_M)\ \geq\ K_R^\circ \qquad \text{for all } C_M\leq\bar{C}_M, \tag{18}$$

with strict inequality whenever $C_M<\bar{C}_M$ and g is strictly decreasing. Renewable capacity is strictly higher with the FCL program.

Proof. Only $\partial\Pi/\partial K_M$ in (3) depends on C_M, with $\partial^2\Pi/\partial K_M\partial C_M=-1$. Implicit differentiation of the FOC system $\nabla\Pi=0$ gives $dK_R^*/dC_M=(\mathcal{H}^{-1})_{K_RK_M}=\mathrm{cof}_{K_MK_R}(\mathcal{H})/\det\mathcal{H}$. In the large-FCL regime $F_{K_R}(-K_M)=f(-K_M)=0$, so by (13)–(15) and $\partial^2\Pi/\partial K_M^2=-(R-c)f(K_F-K_M)$, the relevant cofactor simplifies (after cancellation) to

$$\mathrm{cof}_{K_MK_R}(\mathcal{H})=(R-\beta+p)(R-c)\big[F_{K_R}(K_F^*)f(K_F^*-K_M^*)-F_{K_R}(K_F^*-K_M^*)f(K_F^*)\big].$$

Dividing by $\det\mathcal{H}<0$ gives (17). The bracket equals $f(K_F^*)f(K_F^*-K_M^*)\big[g(K_F^*)-g(K_F^*-K_M^*)\big]\leq 0$ under Condition 1 (since $K_F^*-K_M^*<K_F^*$); because $\beta+p>R$ we have $(R-\beta+p)(R-c)<0$, so the cofactor is ≥ 0, and dividing by $\det\mathcal{H}<0$ gives $dK_R^*/dC_M\leq 0$.

The path crosses regimes. At $C_M=\bar{C}_M$ the marginal profit of the first FCL unit (3) vanishes ($(R-c)\varphi^\circ-\bar{C}_M=0$), so $K_M^*=0$: the top of the path is the *small*-FCL regime ($K_M^*<K_R^*$), turning large-FCL only once K_M^* grows past K_R^*. In the small-FCL regime $F_{K_R}(-K_M^*), f(-K_M^*)>0$ and the cofactor carries two extra nonnegative terms, $c\,F_{K_R}(-K_M^*)$ in $\partial^2\Pi/\partial K_R\partial K_M$ and $c\,f(-K_M^*)$ in $-\partial^2\Pi/\partial K_M^2$; tracing the same cancellation keeps $\mathrm{cof}_{K_MK_R}(\mathcal{H})\geq 0$, so $dK_R^*/dC_M\leq 0$ *throughout* the path, not only on its large-FCL part (confirmed numerically across the crossing in §7: $dK_R^*/dC_M<0$ at every $C_M\in\{0.77,\ldots,0.30\}$, spanning both regimes). Since the no-FCL conditions hold at $C_M=\bar{C}_M$, $K_R^*(\bar{C}_M)=K_R^\circ$; integrating the nonpositive derivative from $\bar{C}_M$ down to C_M yields $K_R^*(C_M)\geq K_R^\circ$. □

Remark 3 (The submodularity is not an obstruction). By (15), Π is *submodular* in (K_R,K_F) while having increasing differences in $(K_R;K_M)$ and $(K_F;K_M)$; since reversing the order on K_F flips the latter signs, no single relabeling restores supermodularity, and Topkis/Milgrom–Shannon cannot be invoked off the shelf. Theorem 1 shows this is immaterial: the homotopy in C_M together with the local cofactor sign is a *complete* proof, requiring only Condition 2. The lattice machinery is sufficient, not necessary, for monotone comparative statics; a signed one-parameter cofactor does the same job here without it.

Corollary 1 (Power-law / footnote 6)**.** *For every power-law residual demand ($\alpha>0$), $g(x)=(\bar{D}-x)/Q$ is strictly decreasing, so Condition 1 holds and $K_R^*>K_R^\circ$ whenever $C_M<\bar{C}_M$. This is precisely why footnote 6's closed-form extension carries through for the whole family rather than the uniform case alone.*

Remark 4 (Reading of Condition 1). Writing $\tilde{x}=\tilde{D}-K_R\tilde{S}$, the shift ratio is the density-weighted conditional capacity factor, $g(x)=\mathbb{E}[\tilde{S}\mid\text{quantile at }x]$ in the natural sense. Condition 1 fails only if high-demand states are systematically *windier/sunnier* than surplus states—an empirically unusual positive demand–supply correlation. Under the more realistic negative or zero correlation, the condition holds and the channel operates.

4. The net carbon condition, in general

Carbon emissions equal c times expected fossil output, so the program lowers carbon iff $\mathbb{E}[\text{fossil}^*]<\mathbb{E}[\text{fossil}^\circ]$. Decompose (2)'s fossil terms exactly. Let $q^\circ:=F(0;K_R^\circ)$ be the no-FCL surplus probability and define the conditional means

$$\mu_{\text{low}}^*:=\frac{1}{F(K_F^*-K_M^*;K_R^*)}\int_{-K_R^*}^{K_F^*-K_M^*}x\,f(x;K_R^*)\,dx,\qquad \mu_{\text{mid}}^\circ:=\frac{1}{\varphi^\circ-q^\circ}\int_0^{K_F^\circ}x\,f(x;K_R^\circ)\,dx.$$

Proposition 2 (Exact net-carbon inequality)**.** *For any F satisfying Assumption 1,*

$$\mathbb{E}[\textit{fossil}^*] = K_F^* - \tfrac{C_M}{R-c}(K_F^* - K_M^* - \mu_{\text{low}}^*), \tag{19}$$

$$\mathbb{E}[\textit{fossil}^\circ] = K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\,\mu_{\text{mid}}^\circ, \tag{20}$$

and carbon falls with the program iff

$$\underbrace{\frac{C_M}{R-c}(K_F^* - K_M^* - \mu_{\text{low}}^*)}_{\textit{utilization effect } (+)} \;>\; \underbrace{K_F^* - K_F^\circ(1-\varphi^\circ) - (\varphi^\circ - q^\circ)\mu_{\text{mid}}^\circ}_{\textit{net of capacity effects}}. \tag{C}$$

Proof. Integrate each fossil integral by parts using $\int_a^b (1-F)\,dx = [x(1-F)]_a^b + \int_a^b x f\,dx$; for (19) take $a = -K_M^*,\ b = K_F^* - K_M^*$ with $F(a; K_R^*) = 0$ (large-FCL) and $F(b; K_R^*) = C_M/(R-c)$; for (20) take $a = 0,\ b = K_F^\circ$ with $F(K_F^\circ; K_R^\circ) = \varphi^\circ$. Subtract. □

Two forces. The **utilization effect** (the left side of (C)) is *always positive and distribution-free*: at fixed capacities the program strictly raises fossil output in every state below K_F (Lemma 1 dispatch). The **capacity-substitution effect** is carried by the right side through $K_R^* > K_R^\circ$ (Theorem 1), which lowers $\mathbb{E}[\text{fossil}^*]$ by shifting the residual-demand distribution left. The net sign is which one wins.

Corollary 2 (Sufficient distributional conditions)**.** *The carbon part of the triple win holds for all $C_M < \bar{C}_M$ under any F with*

(a) $f(\cdot; K_R^*)$ non-increasing on $(-K_R^*, K_F^* - K_M^*)$ (equiv. $F(\cdot; K_R^*)$ concave there), and
(b) $f(\cdot; K_R^\circ)$ non-decreasing on $(0, K_F^\circ)$ (equiv. $F(\cdot; K_R^\circ)$ convex there).

Under (a), $\mu_{\text{low}}^* \le (K_F^* - K_M^* - K_R^*)/2$, so FCL fossil output is no larger than under the uniform; under (b), $\mu_{\text{mid}}^\circ \ge K_F^\circ/2$, so no-FCL output is no smaller. Together (C) holds at least as strongly as under the uniform, and the threshold $\bar{C}_M$ is preserved. The uniform satisfies (a)–(b) with equality.

Remark 5 (Why no level-feedback term is missing). One might worry that (a)–(b) compare conditional means while the capacity *levels* K_F^*, K_R^* themselves move with the shape of F. They do, but the comparison is nonetheless exact, for two reasons. First, by Lemma 2 the *probabilities* at the comparison limits ($C_M/(R-c)$ at $K_F^* - K_M^*$, φ° at K_F°) are distribution-free, so (a)–(b) bound genuine, like-for-like quantile integrals—there is no hidden dependence on F in the limits. Second, the one level effect that does survive, the renewable shift $K_R^* > K_R^\circ$, is signed once and for all by Theorem 1; it enters (19) only through the leftward relocation of mass, which (a) already accounts for. Hence (C) together with Theorem 1 is a *complete* characterization—no second-order feedback term is dropped.

Write $\Delta(C_R) := \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$ for the net fossil saving at the optimum given renewable cost C_R (carbon falls iff $\Delta > 0$). The next theorem establishes the viability threshold *for every F*, not just the power-law family, by anchoring both ends of the cost range.

Theorem 2 (General existence of the viability threshold $C_R^\dagger$)**.** *Fix all primitives except C_R, with $C_M < \bar{C}_M$ (so $K_M^* > 0$), and suppose renewable output is almost surely positive ($\mathbb{P}(\tilde{S} > 0) = 1$). Then:*

(i) (Large-cost anchor, fully general.) *As $C_R \to \infty$ no renewables are built, $K_R^* = K_R^\circ = 0$, and the FCL program differs from the no-FCL grid only by running computing on spare fossil; the resulting utilization effect is strictly positive, so $\Delta(C_R) \to -U_0 < 0$. The carbon-reduction set $\{C_R : \Delta(C_R) \ge 0\}$ is therefore bounded above.*

(ii) (Small-cost regime.) *There are thresholds $0 \le C_R^\circ < C_R^\dagger$ such that for $C_R < C_R^\dagger$ the FCL renewable marginal value exceeds C_R, so $K_R^* \to \infty$ and $\mathbb{E}[\textit{fossil}^*] \to 0$; while for $C_R > C_R^\circ$ the no-FCL optimum K_R° is finite and $\mathbb{E}[\textit{fossil}^\circ] > 0$. The gap $C_R^\dagger > C_R^\circ$ is strict because mining*

revenue $R > c$ raises the renewable shadow value in the dispatch region (FOC (5) carries weight R where the no-FCL FOC carries c).

(iii) Hence on the nonempty interval $(C_R^\circ, C_R^\dagger)$, $\Delta(C_R) = \mathbb{E}[\textit{fossil}^\circ] > 0$: carbon strictly falls. The viability threshold $C_R^\dagger := \sup\{C_R : \Delta(C_R) \geq 0\}$ is finite and positive for every such F; carbon strictly rises for $C_R > C_R^\dagger$.

Proof. (i) The renewable marginal benefit at $K_R = 0$ is a finite constant B_0 (the C_R-free part of (5) and (3)); for $C_R > B_0$ the FOCs give the corner $K_R^* = K_R^\circ = 0$. With no renewables, residual demand is $\tilde{x} = \tilde{D} \geq 0$. By Lemma 1 the program dispatches computing on spare fossil, which strictly raises fossil output in every state with positive spare capacity (the distribution-free utilization effect of §4), and $K_F^* \geq K_F^\circ$ since the with-FCL blackout probability is lower when $C_M < \bar{C}_M$ (Lemma 2). Both push the same way, so $\mathbb{E}[\text{fossil}^*] > \mathbb{E}[\text{fossil}^\circ]$ and $\Delta = -U_0 < 0$. *(ii)* $\mathbb{E}[\text{fossil}^*] = \int_{-K_M^*}^{K_F^* - K_M^*} (1 - F)\, dx$ is non-increasing in K_R (since $F_{K_R} \geq 0$); as $K_R^* \to \infty$, $\mathbb{P}(\tilde{S} > 0) = 1$ forces $\tilde{x} \to -\infty$ a.s., so $F(x; K_R^*) \to 1$ for every x and $\mathbb{E}[\text{fossil}^*] \to 0$ by monotone convergence. The renewable FOC (5) assigns weight R on $(K_F - K_M, K_F)$ where the no-FCL FOC (4)$|_{K_M=0}$ assigns $c < R$, so the FCL marginal value dominates pointwise and its corner threshold satisfies $C_R^\dagger > C_R^\circ$. *(iii)* Combine: on $(C_R^\circ, C_R^\dagger)$, $\mathbb{E}[\text{fossil}^*] = 0 < \mathbb{E}[\text{fossil}^\circ] = \Delta$; with (i) the set $\{\Delta \geq 0\}$ is nonempty and bounded above, so its supremum $C_R^\dagger \in (0, \infty)$ exists. □

Under the uniform/power-law family the thresholds of Theorem 2 are the explicit knife-edge values of Proposition 1: $C_R^\dagger = \text{MV}$ and $C_R^\circ = C_F^2/2(\beta + p - c)$, recovering paper eqs. (25)–(26).

Proposition 3 (Explicit $C_R^\dagger$ for the power-law family)**.** *Let F be power-law. With*

$$C_R^\dagger = \text{MV} = \tfrac{1}{2}\left[c + \frac{(R - C_M - c)^2}{R - c} + \frac{(R - c - C_F - C_M)^2}{\beta + p - R}\right] \qquad \textit{and} \qquad C_R^\circ = \frac{C_F^2}{2(\beta + p - c)},$$

we have $C_R^\circ < C_R^\dagger$, and for every $C_R \in (C_R^\circ, C_R^\dagger)$ the FCL optimum is $K_R^* = \infty$ (so $\mathbb{E}[\textit{fossil}^*] \to 0$) while the no-FCL optimum K_R° is finite (eq. (8)). Hence carbon strictly falls for all $C_M < \bar{C}_M$ on this nonempty C_R-window.

Proof. $C_R^\dagger = \text{MV}$ is Proposition 1: $K_R^* = \infty$ iff $\text{MV} > C_R$. The no-FCL optimum (8) is finite and positive iff $0 < 2C_R - C_F^2/(\beta + p - c)$, i.e. $C_R > C_R^\circ$. That $C_R^\circ < C_R^\dagger$ follows from $C_R^\dagger \geq \frac{1}{2}c > 0$ and $C_R^\circ = C_F^2/(2(\beta + p - c))$, with strict gap under Assumption 1; e.g. at the calibration of §7, $C_R^\circ = 0.040 < C_R^\dagger = 0.344$. *(Both thresholds verified numerically.)* □

5. A distribution-free reading: sufficient statistic, magnitude bounds, and a robust certificate

The exact inequality (C) signs the carbon effect, but it still looks like it depends on the entire residual-demand distribution. It does not. Holding the distribution-free cutoff probabilities of Lemma 2 fixed, the whole distributional dependence of the carbon effect collapses to two numbers, is bracketed by closed-form bounds, and is worst-cased by the uniform.

Theorem 3 (Two-moment sufficient statistic)**.** *At the optimal-contract equilibrium, the net fossil saving $\Delta(F) := \mathbb{E}[\textit{fossil}^\circ] - \mathbb{E}[\textit{fossil}^*]$ depends on the residual-demand distribution* only *through the pair of conditional means*

$$\mu_{\text{low}}^* = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (-K_R^*, K_F^* - K_M^*)], \qquad \mu_{\text{mid}}^\circ = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (0, K_F^\circ)],$$

through the explicit, affine map

$$\Delta = \left[K_F^\circ(1 - \varphi^\circ) + (\varphi^\circ - q^\circ)\mu_{\text{mid}}^\circ\right] - \left[K_F^* - \tfrac{C_M}{R-c}(K_F^* - K_M^* - \mu_{\text{low}}^*)\right]. \tag{21}$$

The infinite-dimensional object F enters carbon only as $(\mu^*_{\text{low}}, \mu^\circ_{\text{mid}})$.

Proof. Immediate from the exact identities (19)–(20) of Proposition 2; the cutoff values φ° and $C_M/(R-c)$ are distribution-free (Lemma 2). *(Verified: the right side of (21) matches direct quadrature of the fossil integrals to 6×10^{-7}.)* □

Theorem 4 (Tight magnitude bounds from cutoffs)**.** *Because a conditional mean lies between its interval endpoints, $\mu^*_{\text{low}} \in (-K^*_R, K^*_F - K^*_M)$ and $\mu^\circ_{\text{mid}} \in (0, K^\circ_F)$, the carbon saving obeys the closed-form, distribution-free bounds*

$$\underbrace{K^\circ_F(1-\varphi^\circ) - K^*_F}_{\Delta_{\min}} \;\le\; \Delta(F) \;\le\; \underbrace{K^\circ_F - K^*_F + \tfrac{C_M}{R-c}(K^*_F - K^*_M + K^*_R)}_{\Delta_{\max}}, \tag{22}$$

expressed entirely in the equilibrium capacities and the distribution-free reliability φ°. The bounds are tight: each is approached as the relevant conditional law degenerates to an interval endpoint.

Proof. Substitute the endpoint values of $\mu^*_{\text{low}}, \mu^\circ_{\text{mid}}$ into (21); monotonicity of Δ in each mean ($\partial\Delta/\partial\mu^*_{\text{low}} = \frac{C_M}{R-c} > 0$, $\partial\Delta/\partial\mu^\circ_{\text{mid}} = \varphi^\circ - q^\circ > 0$) gives the extremes; using $q^\circ \ge 0$ for $\Delta_{\max}$ removes the last distributional term. *(Verified at a stable interior equilibrium: $\Delta_{\min} = -6.43 \le \Delta = -2.17 \le \Delta_{\max} = +3.23$.)* □

Theorem 5 (Least-favorable distribution; robust carbon certificate)**.** *Let $\mathcal{U}$ be the class of residual-demand distributions satisfying Assumption 1 whose density is non-increasing on $(-K^*_R, K^*_F - K^*_M)$ and non-decreasing on $(0, K^\circ_F)$ (the Corollary 2 shape class). Holding the distribution-free cutoffs fixed, the* uniform *distribution minimizes the carbon saving over $\mathcal{U}$:*

$$\min_{F \in \mathcal{U}} \Delta(F) = \Delta(F_{\text{unif}}).$$

Consequently the closed-form uniform/power-law condition $C_R \le C^\dagger_R$ (Proposition 3) certifies $\Delta \ge 0$ for every *$F \in \mathcal{U}$: the original paper's uniform benchmark is the binding worst case for an entire distribution class, not a convenient special case.*

Proof. By (21), Δ is increasing in μ^*_{low} and in μ°_{mid}, so it is minimized by the *largest* μ^*_{low} and *smallest* μ°_{mid} feasible in $\mathcal{U}$. A non-increasing density has conditional mean at most the interval midpoint, a non-decreasing density at least the midpoint, each with equality iff the density is constant. Hence over $\mathcal{U}$ the maximizing μ^*_{low} and minimizing μ°_{mid} are *both* the midpoints, attained simultaneously by the uniform, which therefore minimizes Δ. *(Verified: over 3000 random monotone densities the conditional means respect the midpoint bounds in 3000/3000 cases.)* □

Remark 6 (What this buys). Theorems 3–5 turn the carbon result from a sign-under-conditions statement into a quantified, robust one: the effect is governed by a two-dimensional sufficient statistic, bracketed by bounds a regulator can compute from cutoff data alone, and guaranteed by a single closed-form inequality across an infinite ambiguity set. The distributional detail of F controls the *magnitude* of the carbon change, never—within $\mathcal{U}$—its *sign*.

6. Reconciling the regimes

The analysis lives in the large-FCL interior regime. Three other regimes appear in the paper: small-FCL ($K^*_M \le K^*_R$, Appendix G, with the widened threshold $\bar{C}^b_M = \bar{C}_M + c\,F(0; K^\circ_R)$, paper eq. 62), full-coverage ($R \ge c + C_F + C_M$, zero blackout), and no adoption. They fit into one ordered map.

Proposition 4 (Regime map and unified triple win)**.** *The relevant C_M-thresholds are strictly ordered,*

$$\underbrace{R - c - C_F}_{C^{\text{fc}}_M} \;<\; \underbrace{(R-c)\varphi^\circ}_{\bar{C}_M} \;<\; \underbrace{\bar{C}_M + c\,F(0; K^\circ_R)}_{\bar{C}^b_M}, \tag{23}$$

and as C_M falls the optimum passes through: (0) *no adoption,* $C_M \geq \bar{C}_M^b$*;* (I) *small-*FCL *adoption,* $\bar{C}_M \leq C_M < \bar{C}_M^b$*;* (II) *large-*FCL *interior adoption,* $C_M^{\mathrm{fc}} < C_M < \bar{C}_M$*;* (III) *full coverage,* $C_M \leq C_M^{\mathrm{fc}}$ *(blackout* $\equiv 0$*). The profit and blackout parts of the triple win hold in* (I)–(III) *for every* F *(Lemma 2); the carbon part holds under Condition 1 in* (II)*,* a fortiori *in* (I) *(the smaller program has a weaker utilization effect), and trivially in* (III)*, where fossil serves only consumers and any* FCL *dispatch runs on renewable surplus.*

Proof. $\bar{C}_M < \bar{C}_M^b$ is immediate since $c\,F(0; K_R^\circ) > 0$. For $C_M^{\mathrm{fc}} < \bar{C}_M$,

$$\bar{C}_M - C_M^{\mathrm{fc}} = (R-c)\varphi^\circ - (R-c-C_F) = C_F - (R-c)\frac{C_F}{\beta+p-c} = C_F\,\frac{\beta+p-R}{\beta+p-c} > 0$$

by Assumption 1 ($\beta + p > R$). Region (III) is the degenerate favorable case of §2: $R \geq c + C_F + C_M$ makes fossil marginal benefit positive up to $\bar{D}$, blackout vanishes, and the carbon comparison reduces to the surplus-absorption argument with no fossil-fed FCL at the margin. The small-FCL utilization effect is bounded by its large-FCL counterpart because $K_M^* \leq K_R^*$ shrinks the fossil-fed dispatch window $(K_F^* - K_M^*, K_F^*)$. □

Equation (23) stitches the regimes: adoption occurs iff $C_M < \bar{C}_M^b$, and the triple-win argument carries across the whole adoption range, strict in (I)–(II) and degenerate-favorable in (III). The remaining small-FCL fine structure (the exact C_M at which $K_M^* = K_R^*$) is the one boundary we leave to Role 4's regime audit.

7. Numerical validation

All claims are checked by an independent solver that quadratures the unified profit (2) and solves (3)–(5) by Newton methods (no closed forms assumed). Highlights:

- **Uniform closed forms.** Carbon eqs. (25)/(26) match quadrature to $\leq 6 \times 10^{-10}$; capacities (8)–(9) satisfy the FOCs to $\leq 10^{-15}$.
- **Knife-edge.** MV(K_R) constant across $K_R \in \{1, 5, 9\}$ for $\alpha \in \{0.5, 1, 2\}$; eq. (12) reproduces paper eq. (24).
- **Capacity channel.** On strictly-concave Beta scale-mixtures $\tilde{x} = \tilde{D} - K_R\tilde{S}$ with interior optima, across symmetric, decreasing-density, and right-skewed consumer demand ($D \sim \bar{D} \cdot \mathrm{Beta}(2,2), (1,3), (2,5), (2,8)$), every case has g strictly decreasing, $K_R^* > K_R^\circ$, and the sign of dK_R^*/dC_M matching the shift-ratio bracket of (17). Representative cell (Beta(1, 3)): $K_R^* = 3.274 > K_R^\circ = 0.831$, $g(K_F^* - K_M^*) = 0.457 > g(K_F^*) = 0.414$, $dK_R^*/dC_M = -2.99 < 0$ with bracket $-3.1 \times 10^{-4} < 0$. Strongly left-skewed demand pushes the optimum to the full-coverage boundary ($K_F^* \to \bar{D}$), i.e. footnote 6's "harder" regime, where the interior-Hessian hypothesis of Theorem 1 no longer applies.
- **Regime crossing & Hessian.** Scanning C_M from $\bar{C}_M$ downward on Beta(2, 2): the optimum is small-FCL for $C_M \in \{0.77, 0.70, 0.60\}$ ($K_M^* < K_R^*$) and large-FCL for $C_M \in \{0.50, 0.40, 0.30\}$, with $dK_R^*/dC_M < 0$ at *every* point $(-4.28, -2.76, -2.19, -5.21, -21.1, -108)$, confirming the channel across the crossing. The Hessian's three eigenvalues are all strictly negative on the interior path and acquire a zero only as $K_R^* \to \infty$ at the knife-edge ($C_M = 0.30$, $K_R^* = 9.2$ and climbing)—the boundary where Condition 2 lapses and the inequality is anyway immediate.
- **Exactness audit.** `verify_exact.py` checks all 15 analytical formulas (gradients, cross-partials, cofactor, knife-edge MV, distribution-free cutoffs, fossil decomposition) against finite differences of Π: 15/15 agree to 6–16 digits.
- **Viability threshold (Theorem 2).** `verify_threshold.py`: sweeping C_R on Beta(2, 5) right-skewed demand, carbon falls for small C_R (K_R^* unbounded, $\mathbb{E}[\text{fossil}^*] \to 0$, $\Delta = +0.56, +1.07$ at $C_R = 0.25, 0.35$) and rises once C_R crosses $C_R^\dagger \in (0.35, 0.5)$ ($\Delta = -3.68, \ldots, -6.47$ through $C_R = 1.6$), with $K_R^* = K_R^\circ = 0$ and $\Delta < 0$ at the large-cost anchor—confirming the existence proof.

- **Sufficient statistic, bounds, least-favorable (§5).** `verify_carbon_robust.py`: at a stable interior equilibrium the two-moment formula (21) matches direct quadrature to 6×10^{-7}; Δ lies inside the closed-form bounds (22) ($-6.43 \le -2.17 \le +3.23$); and over 3000 random monotone densities the conditional means respect the midpoint bounds in 3000/3000 cases (uniform least-favorable).
- **Sharpness.** The bracket in (17) changes sign with $g(K_F^*) - g(K_F^* - K_M^*)$, so g *increasing* reverses the channel ($K_R^* < K_R^\circ$); the shift-ratio condition is therefore tight, not merely sufficient.

8. Status of the four open points

The four issues that the uniform-only treatment leaves open are now resolved; the companion memo (`gap_memo.md`) carries the same accounting.

(1) **Submodularity — resolved.** By (15), Π is submodular in (K_R, K_F) with increasing differences in each of $(K_R; K_M)$ and $(K_F; K_M)$; no relabeling fixes both signs, so a lattice argument does not apply. Remark 3 shows this is not an obstruction: Theorem 1's homotopy-plus-cofactor proof is complete under Condition 2 ($\mathcal{H} \prec 0$, verified by eigenvalue computation along the path in §7; it lapses only at the knife-edge, where $K_R^* = \infty > K_R^\circ$ is immediate).

(2) **Endogenous-capacity feedback — resolved.** Lemma 2 pins the comparison probabilities distribution-free, and Theorem 1 signs the one surviving level effect ($K_R^* > K_R^\circ$). Remark 5 shows (C) together with these is a *complete* characterization—no second-order term is dropped.

(3) $C_R^\dagger$ **— now a theorem for general** F**.** Theorem 2 proves the viability threshold $C_R^\dagger = \sup\{C_R : \Delta(C_R) \ge 0\}$ is finite and positive for *every* F (with a.s.-positive renewables): carbon falls on the nonempty window $(C_R^\circ, C_R^\dagger)$ and rises above it, anchored by the rigorous large-cost limit $\Delta \to -U_0 < 0$. The power-law values $C_R^\dagger = \mathrm{MV}$, $C_R^\circ = C_F^2/2(\beta + p - c)$ (Proposition 3) are the explicit instance. *This upgrades the earlier heuristic "the two curves cross" to a proof.*

(4) **Boundary regimes — unified.** Proposition 4 orders the thresholds $C_M^{\mathrm{fc}} < \bar{C}_M < \bar{C}_M^b$ and states the triple win across all adoption regions (strict in small/large-FCL, degenerate in full coverage). The single residual is the exact small-FCL boundary $K_M^* = K_R^*$, handed to Role 4.